

Unpacking Children's Perceptions of Their Digital Money Futures

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Abstract

Children's encounters with money are increasingly shaped by cashless, platform-based, and gaming systems, however limited HCI research examines how children experience and negotiate these changes. This paper presents preliminary findings from a mixed-method, participatory design futuring study with children aged 10–11 in a UK primary school, a key stage when children typically shift from cash to digital money, and experience growing financial autonomy. Conducted over a period of 5 weeks, the study involved surveys, workshops, focus groups, and interviews, exploring children's experiences of multiple economies and visions for future financial systems. Initial findings show that children actively navigate cash, digital money, and in-game currencies, valuing visibility, safety, and control, while expressing concerns about scams and financial harm, particularly in games. Through speculative 'super-inventions,' children's work revealed tensions between protection and autonomy, highlighting opportunities for designing financial technologies that would better support and educate children during this critical transition.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; **Empirical studies in HCI**.

Keywords

Children's Perceptions, Child-Computer Interaction, Money, Digital Money, Futuring Workshops, Surveys

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1 Introduction & Related Work

In recent years, the HCI community has identified the changing nature of how children understand, learn about and experience money, as it is materialised and encountered in cashless, networked, digital forms [1, 12, 13]. Children's financial encounters may now simultaneously include cash, card payments, gift cards, platform and gaming currencies (e.g. *V-Bucks*, *Robux*), pocket money apps (e.g. *GoHenry*, *Rooster*, *HyperJar*), cryptocurrencies, children's bank accounts and longer-term investments or savings supported by family members. Children participate in a digital economy, with their peers, and alongside parents, through advertising, online shopping, second-hand selling and increasingly spend money on virtual goods (e.g. game items, online subscriptions, platform currencies). In the UK, 71% of children aged 7–17 pay for things online themselves and 53% of those playing games have spent money online within these games [9]. In a review of children's digital financial literacy, Lukas et al. [7] suggest possible benefits to digital financial experiences, including: new earning and learning opportunities, the financial inclusion and improving financial management. However, they also identify significant challenges and risks for young people including increased exposure to online scams and fraud, impulsive spending and gamified financial habits.

There have been a small number of significant recent HCI papers devoted to children and young people's experiences of digital monies. In 2023, Yip et al. [13] described a participatory design study with children aged 7–11, reflecting on the role of technology in their financial lives, and financial literacy in particular. Andries et al. [1] presented a cultural probe study, with children (aged 7–12) and their families, seeking to understand the 'secret lives of children's digital monies' and the ways their conceptions and lived experiences of money may be changing. Other scholars have devoted attention to the implications of monetisation in games and platforms, with Hardwick et al. [5] in particular identifying how children (aged 8–13) recognised and understood harms they experienced when spending and managing money within the popular game Roblox. As a brief summary, recurrent issues identified in

this prior work include the ‘moneywork’ [10] and management involved for children and parents; the ongoing significance of cash for younger children, in increasingly cashless societies; the promise and perils of financial interactions and monetisation in games, as well as children’s sophisticated understanding of value and harm in these contexts. These works also reflect on the kinds of financial education and tools that children, young people and their parents require to navigate ‘new money’ [12] as well as the tensions raised between supporting and protecting young people when interaction with digital money, while enabling them to develop financial capability and autonomy through independent experiential learning. Most of these prior studies have been exploratory, or focused on a specific context (e.g. gaming), or goal, (e.g. financial literacy).

In the UK context, 10–12 is a key transitional age, as children are on the cusp of moving from primary school (ages 5–11) to secondary school (ages 12–18). Most traditional children’s bank accounts in the UK begin at age 11; 91% of children in the UK have access to their own mobile device at this age [8]. However, there is limited research which has taken place with children at this specific age, to understand their lived experiences of money, and their aspirations for how money and financial services could be better designed for them in the future. In this paper, we describe early findings from a large mixed method, participatory design and futuring engagement (including surveys, focus groups, interviews and design sessions) with 57 children (aged 10–11) at a UK primary school over 5 weekly sessions. Our study demonstrates children’s diverse experiences and future perspectives at this key transitional age.

2 Methodology

Our primary aim was to enable children to engage with, reflect upon, and then design possible futures of digital money. We hoped to understand their *present* experiences, attitudes and concerns, as well as surface their perceptions of possible *futures* of money, such as different ways to access and manage digital money without a smartphone, learning about money in a wholly cashless society, or what rules or systems they wished applied to in-game financial transactions. We employed a mixed-methods approach, engaging primary-school children in the UK in a survey and series of design futuring workshops. A total of 57 participants (ages 10–11) with mixed socio-economic backgrounds, from two classrooms at Primary 7 (P7) level, were recruited in partnership with a state-run primary school near a large UK city. The methodology and content were discussed in detail and agreed with teaching staff. The study was ethically reviewed and approved by the University. Participant information sheets and consent forms were provided for both parents and children. Ongoing assent was actively sought and monitored by the research team at each workshop to ensure that participation remained voluntary throughout.

2.1 Study Design

Our study first asked children to respond to a take-home survey. This survey was designed based on prior exploratory research, and provided broad background information about the participants and their initial perspectives on money in their lives. The survey was then followed by four workshops, which employed participatory methods regularly used in HCI research with children (e.g. [2, 4, 11])

alongside a Participatory Design Futuring approach [3]. Principles of participatory design (centering children’s perspectives on how technologies are designed for them) are combined with ‘design futuring’ [6], where design artefacts or scenarios are co-created, to enable discussion of possible futures. We ran four design workshop sessions, for 90 minutes, over four weeks, with two P7 classes, supported by the class teacher. The children worked in consistent, mixed ability teams of 3 to 4 pupils, and were provided with various bespoke designed paper worksheets for each session. Activities were framed as ‘missions, which the research team required pupils help to complete. Children were invited to play different roles (detectives, inventors, storytellers or designers).

A detailed description of the design of these workshops is beyond the scope of this work-in-progress paper. In summary, in **Workshop 1** children played the role of ‘detectives’ to explore literature and media focused on a range of themes derived from previous research, presented as ‘evidence packs’ in a child friendly format. Themes revolved around: accessing (digital) money; managing (digital) money in the future; tackling in-game money and currencies; and learning about money in a cashless society. This provided children with a background to the research topic, introduced them to known challenges related to the changing nature of money in children’s lives and invited them to reflect on their own financial lives and practices. In **Workshop 2** the children worked both individually and in teams to ideate future services and products that could address a challenge they had identified from the evidence in the previous workshop. In **Workshop 3**, each team (15 groups total) refined and combined these ideas to produce one ‘super-invention’ for digital money in the future. Each team pitched their ideas to the rest of the class, through an A1-sized worksheet that allowed them to present their idea, the target audience, the pros and cons, as well as advertisement strategies, using multi-modal visual aids such as their own drawings and cutouts from magazines. In **Workshop 4** each child individually wrote a short story about when, where and how that super-invention may be used in practice. In Week 5, a selection of children participated in focus groups, or one-to-one interviews to share and reflect on their individual stories.

2.2 Data Collection & Analysis

Alongside the workshops, we collected data in multiple forms, including written surveys, in-class audio recordings, focus groups, and brief 1-1 interviews. Two surveys were distributed, one prior to the workshops, and another survey afterwards. The survey data was digitised for analysis. Workshop materials produced from each class activity were photographed, and where relevant class presentations were audio recorded. All audio was professionally transcribed. At the time of writing, qualitative analysis of all survey, interview and workshop data is ongoing. For this work-in-progress paper, we present an initial descriptive analysis of the survey results, as well as a descriptive account of the various workshop outcomes, with examples of how children approached the design activities, and primary points of discussion.

3 Initial Findings

3.1 Survey Responses

We gathered 57 responses to the initial survey. Not every child responded to every question and as such we report our findings through relative frequencies. To provide a brief overview, 45/57 participants, have their own smartphone, 33 children reported having a bank account, and 28 children reported having access to a bank card. The survey surfaced numerous findings and areas of interest which we briefly present below.

3.1.1 Understanding of Core Financial Concepts. Children at this age demonstrated a clear understanding of the core concepts of money. When asked how they would explain what money is to an alien, their answers revolved around three themes. The first was **what money can provide**. Children reflected on money's value both in terms of needs (e.g. *"money is a thing to help you buy food or could get you warm clothes"*) and wants (e.g. *"it buys you your favourite stuff"*). The second was the **core transactional function of money**. For example, one participant described that *"money is currency that human use like an exchange."* Finally, the children also sometimes considered the **form money can take**, for example, describing money in terms of *"paper and metal tokens that you can buy items with."*

3.1.2 Participating in a mixed currency economy, including cash. We also sought to understand the children's views on different currencies, including cash, bank accounts, game credits, and cryptocurrencies. When asked which medium is 'best' for holding £20, responses were split between cash (22/55) and a bank account (24/55), showing the ongoing significance of cash at this age. The majority (36/55) perceived money kept in a bank account as the safest option due to its security features and the oversight functionality for parents. For example, children reflected that *"bank accounts have a lot of passwords and personal info to get in"*, and that they are safer *"because my parents can see it."* These findings reveal that children exhibit diverse preferences regarding different forms of money even at a relatively young age.

3.1.3 Gaming offering entrepreneurship and education while creating harms. Over half of the children who responded spent money in games (n=32/45), 5 children generating income within these games and 6 reporting trading virtual items. Strikingly, 'game credits' were a preferred currency for 13 children, but few reported that this was safest. This reveals the importance gaming plays in children's understanding of financial concepts by offering an immersive setting for learning about value exchange, trading, spending, and entrepreneurship. When asked to share instances of negative experiences with money in games, almost a third of participants reported being scammed or described situations of manipulation, e.g. *"I bought something that I didn't want"* showing the potential for financial harm through games.

3.2 Overview of Children's 'Super-Inventions'

The workshops resulted in a total of 15 'super-invention' ideas, 8 from the first class and 7 from the second, derived from three weeks of group exploration, ideation and discussion. We summarise these

here to provide indicative findings of what money trends, concepts and design ideas the children found most relevant or compelling.

3.2.1 'Super-invention' Example: "Techify". As an illustrative example of a 'super-invention', we discuss in more depth an app created by one of the teams, entitled 'Techify' (see Figure 1). Working within the theme of 'managing' money, their proposed application was designed to promote safer spending practices among children by embedding parental oversight and control mechanisms. This was intended to function across all applications which might enable or encourage spending, including games. The application would require authorisation through a PIN known only to a parent or guardian, or through biometric verification such as fingerprint or Face ID, thereby preventing excessive or unauthorised purchases. This idea emerged from children's earlier reflections, particularly concerns about the ease with which young people can make purchases on shared family devices. As one participant later noted during the focus group, children can *"just click and buy"* repeatedly, often unintentionally, referring to younger users who, in their view, may lack the understanding or self-regulation required to navigate digital marketplaces responsibly. The app reflects children's recognition of the importance of learning financial responsibility as well as the role of technical safeguards and adult supervision to support children achieving this.

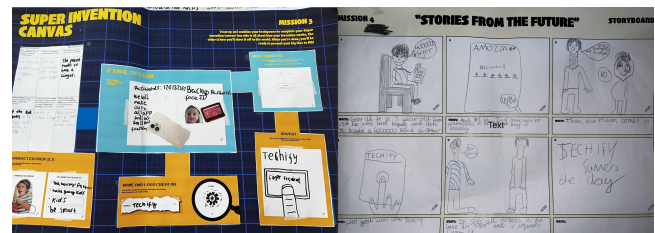


Figure 1: The image on the left depicts a 'super-invention' board, showing the work of group 3 from class 2 (Workshop 3) - 'Techify'. The image on the right shows the individual work of one of the children from group 3 done in storyboard format (Workshop 4)

3.2.2 Broader findings. The majority of the 'super-inventions' depicted mobile applications (apps) for accessing, managing or learning about money, irrespective of the initial themes that the teams were provided (7 in first class; 4 in second class). This may be due to the high number of pupils who own or have access to a smartphone, the familiarity of the 'app' format as a regularly used or encountered artefact, or simply as a tool that children appear to enjoy using in general, as discussed by some of the participants in the focus groups.

In terms of content, a significant amount of these ideas appeared to include or require a variety of safety protocols for accessing the respective apps, ranging from PIN codes to Face ID or fingerprints, irrespective of the age that their target users were supposed to be (ranging from very young children aged 5 to pre-teens and teenagers). The intentions behind these ideas seemed to reflect the need to protect children (as technology users) from financial scams, while highlighting the necessity or expectation of parental

oversight in children's access and use of digital forms of money, which resonates strongly with perspectives expressed in the survey findings.

Children's imaginaries of the future of finance further illustrated a critically reflexive engagement with aspects such as digital monetisation, financial management, and accessibility challenges. Their ideas highlight children's concerns about being protected from commercial pressure, particularly in digital gaming environments where monetisation strategies may undermine their autonomy. Furthermore, some children's designs reflect their thoughts about financial transparency and learning, positioning the visibility of money as central to developing financial competence and autonomy. Attention was also paid by the children to accessibility and inclusion, recognising how existing financial infrastructures can marginalise or expose vulnerable children to risk, such as visually-impaired users. For example, one team developed the "*Bank of Braille*", an inclusive banking concept designed to support people with partial or complete vision impairment in carrying out financial transactions independently. Their super invention was envisioned as a cash machine operating similarly to a conventional ATM, but what made it stand out was its accessibility orientation, as it would support a Braille interface while also including AI-powered voice guidance that "*speaks to you and helps you.*"

4 Discussion & Conclusion

4.1 Designing for Mixed Economies and Money Transitions

Although beginning with four core themes (accessing money, managing money, learning about money and gaming money), as a highly open-ended set of activities, our research highlights how mixed, transitional and multi-faceted children's relationships with money are at the age of 10-11 years. As highlighted in prior work by [1] transitioning from cash to digital money is now a key aspect of growing up and gaining financial autonomy. However, it remains striking in the context of a 'cashless society' how many children at this age remain 'cash-only'. In the short-term, it is a considerable challenge to design for this plurality, ensuring children and families who continue to primarily use cash are not financially excluded, while simultaneously supporting children as they encounter various forms of digital monies at younger ages. Children's designs expressed the need for digital money to be more visible and accounted for, beyond simply viewing a balance on a screen. Key areas for further work include **designing for this particular financial transition** as a key step in children's development (with more support for parents to enable it). This entails not just the design of apps or financial technologies, but programmes for financial education. We believe there is also an opportunity for phy-gital devices, enabling diverse and physical representations of digital money (e.g. digital piggy banks). Similarly, touchpoints and services where children exchange currencies, where children can see and recognise when money changes form, also seems a significant opportunity for further design research.

4.2 Overcoming Fears of Scams and Negotiating Financial Harm

Across all of the futures children produced, we were particularly struck by the many fears and concerns expressed by children about being scammed. There are many possible explanations for this: scams are often in the news, and something they are likely warned a lot about by parents and schools; or they may be channeling what they think adults or teachers would want to hear. Nonetheless, from the survey, it is clear even at this age children already have very real experiences of financial harm, especially through the monetisation of games. Recent research in the context of Roblox by [5] demonstrates how sophisticated children can be in identifying and expressing financial harm. Several children's designs in our study implied futures where technology could be used in a highly surveillant way in order to protect them from potential harm (although not fully relying on platforms themselves for user safety). Carefully considered design is required to ensure children's financial safety, while developing financial autonomy and confidence. It is also important to understand and support constructive conversations between children and responsible adults about these significant experiences (including educating parents about the kinds of financial harm children may encounter in games).

5 Conclusions and Further Work

Though still at an early stage, our findings show that children aged 10–11 navigate the financial landscape through mixed currencies and transitional experiences, where cash, digital money, and other currencies have to coexist. Designing for this stage requires supporting multiple monetary practices, easing the transition from cash to digital through visible and phy-gital representations, and addressing fears of scams while fostering autonomy. Our further data analysis will examine and articulate in greater detail the breadth of possible financial futures children envisage and elicit further reflection on these from both our original participants, and wider stakeholders.

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